

THE PROBLEM

HealthConnect Clinic misses close to half of its scheduled appointments, wasting slots and disrupting care. This model predicts, before an appointment happens, how likely a patient is to miss it — using information already collected at booking time.

MODEL PERFORMANCE

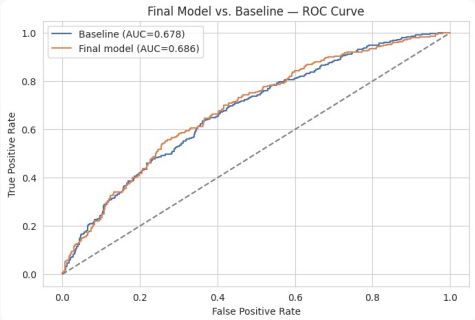
0.686

ROC-AUC FINAL

0.677

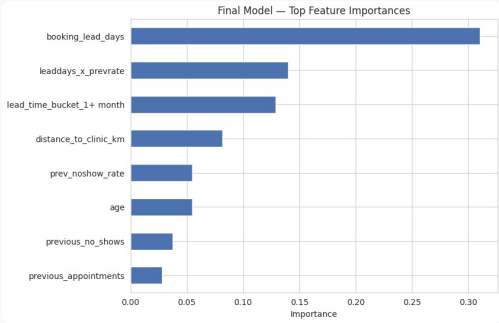
ROC-AUC BASELINE

Random Forest, cross-validated stable at 0.677 (± 0.008) across 5 patient-grouped folds — the improvement over the Week 5 baseline is real, not a lucky split.



WHAT DRIVES RISK

Booking lead time and its interaction with prior no-show history dominate. Distance to the clinic ranks high too — see the insight below.



BUSINESS VALUE: PRIORITISED OUTREACH

72%

PRECISION, TOP 20% RISK

50%

BASE NO-SHOW RATE

29%

NO-SHOWS CAPTURED

Flagging just the riskiest 20% of appointments for extra outreach hits genuine no-shows 72% of the time — far better than treating every patient the same.

KEY INSIGHT: REMINDERS AREN'T ENOUGH FOR DISTANT PATIENTS

Reminders cut no-show rates by ~10 points for mid-distance patients, but only ~2 points for the farthest quintile. A text/WhatsApp nudge alone doesn't work for patients far from the clinic — they likely need a different intervention (telehealth, transport support, or a different scheduling approach).

✓ Use it to

- Prioritise a limited outreach budget
- Flag patients needing a different intervention

✗ Don't use it to

- Auto-cancel or auto-overbook slots
- Replace clinical/admin judgment

HANDOFF TO ML ENGINEERING

Input: 11 raw appointment fields (Data Dictionary) + 4 engineered features computed from them. **Output:** a risk probability (0-1), meant for ranking, not a hard cutoff.

Known issue: an unseen category (e.g. a new reminder channel) is currently silently absorbed rather than flagged — needs explicit validation before production use.